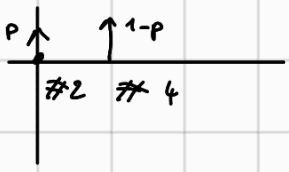
feature vektörü

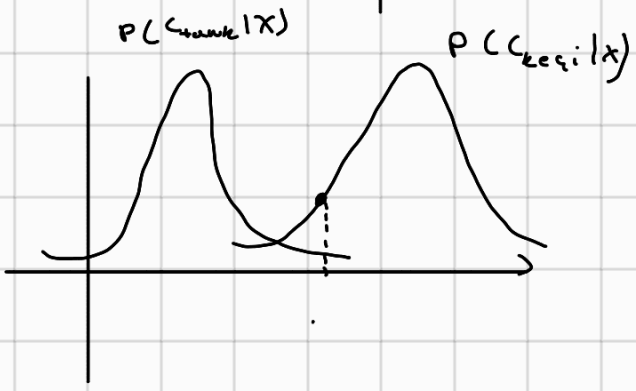
$$\vec{x} = \begin{bmatrix} \#\text{ayak sayısı} \\ \#\text{ağırlığı} \end{bmatrix}$$

$P(C_i|x)$

tavuk, kedi } #ayak ayısı $r^t \in \{2, 4\}$

Benolli dağılımı :





Parametric kestirim:

① Varsayılan Model : θ

② En olabilirlik fonksiyonu : $L(\theta)$

③ Optimization Yöntemi : MLE maximum likelihood estimation

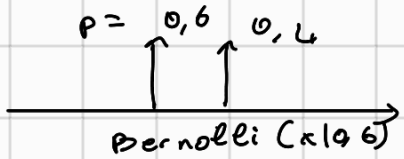
MLE:

$$X = \{x^1, x^2, \dots, x^N\}$$

her bir örnek

$x^t \sim p(x|\theta)$

bu dağılıma göre üretilmiş



$$L(\theta|X) = p(x^1|\theta) \cdot p(x^2|\theta) \cdots$$

$$= \prod_{i=1}^{N} p(x^i|\theta)$$

0'dan büyük

$\hat{\theta} = \underset{\theta}{\text{argmax}}\ L(\theta|x)$ = likelihood fonksiyonunu max yapan θ benim kestirim

logaritmasını alabilirim

$$\ell(\theta|x) = \log L(\theta|x)$$
$$= \sum_{i=1}^{N} \log P(x^i|\theta)$$

Yeterli istatistik (Sufficient Statistics)

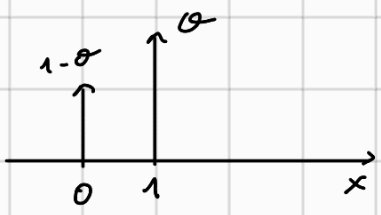
Bir dağılımı açıklayabilecek minimum parametre sayısı

Bernoulli Distribution:

$$P(x|\theta) = \theta^x (1-\theta)^{1-x}, \quad x \in \{0,1\}, \quad \theta \in [0,1]$$

(x: 0 ya da 1; θ: 0 ile 1 arasında)

??



$$L(\theta|X) = \prod_{i=1}^{N} P(x^i|\theta)$$
$$= \prod_{i=1}^{N} \theta^{x_i} (1-\theta)^{(1-x_i)}$$
$$\ell(\theta|X) = \sum_{i=1}^{N} \left[(x_i \log\theta) + (1-x_i) \log(1-\theta) \right]$$

$x_i \in \{0,1\}$

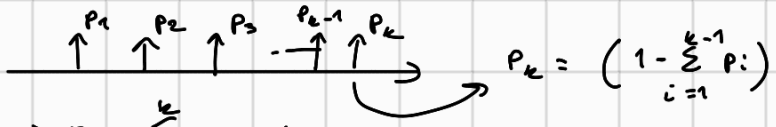
$$S = \sum_{i=1}^{N} x_i$$

$$\frac{\partial \ell(\theta|X)}{\partial \theta} = 0 = \frac{\partial (S \log\theta)}{\partial \theta} + \frac{\partial (\log(1-\theta)(N-S))}{\partial \theta}$$

$$= \frac{S}{\theta} - \frac{N-S}{1-\theta}$$

$$\frac{N-S}{1-\theta} = \frac{S}{\theta}$$
$$S - S\theta = N\theta - S\theta$$
$$\hat{\theta} = \frac{S}{N} = \frac{\sum_{i=1}^{N} x_i}{N}$$

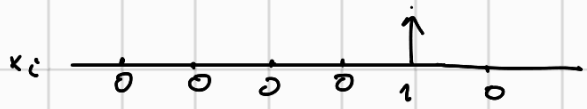
Saç renkleri üstünden konuşalım



$$p_k = \left(1 - \sum_{i=1}^{k-1} p_i\right)$$

$$p_i \geq 0, \quad \sum_{i=1}^{k} p_i = 1$$

Classlarım var sınıf vektörü



← Multinomial Density

$$P(x|\theta) = \prod_{k=1}^{K} p_k^{x_k}$$

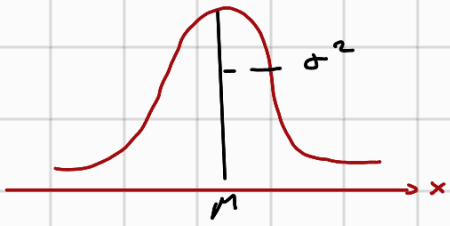
$$L(\theta|\mathcal{X}) = \prod_{i=1}^{N} P(x_i|\theta) = \prod_{i=1}^{N} \prod_{k=1}^{K} p_k^{x_i}$$

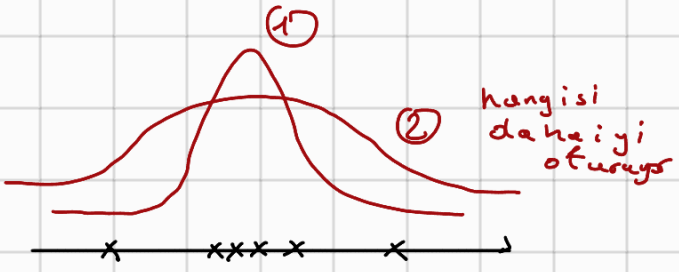
$$l(\theta|\mathcal{X}) = \log \left(\prod_{i=1}^{N} \prod_{k=1}^{K} p_k^{x_i} \right)$$

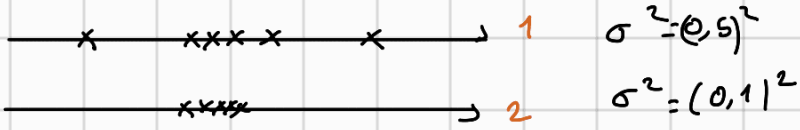
$$\frac{\partial\, l(\theta|\mathcal{X})}{\partial\, p_1} = 0 \Rightarrow \hat{p}_1 = \frac{s_1}{N}$$

Gauss Dağılımı

$$p(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$







$$L(\theta|\mathcal{X}) = \prod_{i=1}^{N} P(x_i|\theta)$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^N \prod_{i=1}^{N} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$l(\theta|\mathcal{X}) = -N \log(2\pi\sigma^2) - \sum_{i=1}^{N} \frac{(x-\mu)^2}{2\sigma^2}$$

$$\frac{\partial\, l(\theta|\mathcal{X})}{\partial \mu} = \frac{1}{2\sigma^2} \sum_{i=1}^{N} 2(x_i - \mu)$$

$$0 = \frac{\sum_{i=1}^{N} x_i - N\mu}{\sigma^2}$$

$$N \cdot \mu = \sum_{i=1}^{N} x_i$$

$$\boxed{\hat{\mu} = \frac{1}{N} \sum_{i=1}^{N} x_i}$$

$$\hat{\mu}_{ML}^{T} = \frac{1}{T} \sum_{t=1}^{T} x_t = \frac{1}{T} \left(x_t + \sum_{\tau=1}^{T-1} x_\tau \right) = \frac{x_t}{T} + \frac{1}{T} \underbrace{\sum_{t=1}^{T-1} x_t}_{\hat{\mu}_{ML}^{(T-1)}}$$

↓ Maximum likelihood

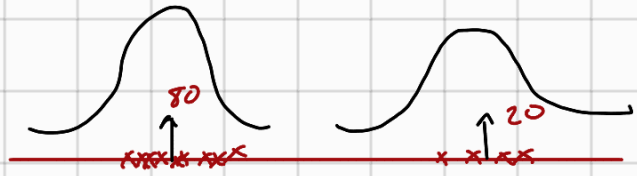
$$= \frac{x_t}{T} + \frac{1}{T} \hat{\mu}_{ML}^{(T-1)} (T-1)$$

$$\hat{\mu}_{ML}^{T} = \frac{1}{T} x_T + \frac{T-1}{T} \hat{\mu}_{ML}^{T-1}$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{N} (x_i - \mu)^2 = 0$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{\mu})^2$$

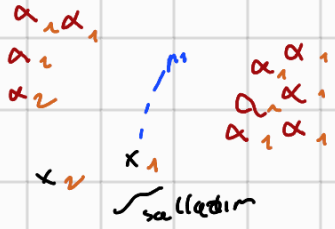
Gaussian Mixtures



$$P(x) = \sum_{k=1}^{K} w_k \, \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

↓ kovarians matrisi

k-means:



$w_k = ?$
$\mu_k = ?$
$\Sigma_k = ?$

EM algorithm

Expectation maximization algorithm

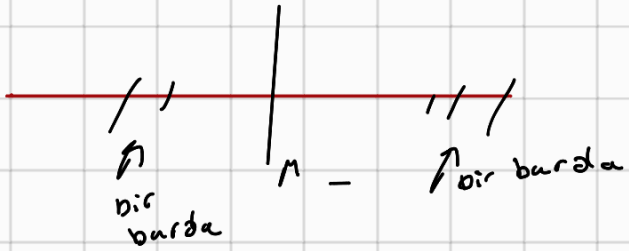
$$\sum_{k=1}^{K} w_k^0 \, \mathcal{N}(\mu_k^{(0)}, \Sigma_k^{(0)}) = P(x \mid \theta)$$

↳ Etiketlere Expectation

↳ Maximization → $\sum_{k=1}^{K} \mu_k^{(1)} \, \mathcal{N}(\mu_k^{(1)}, \Sigma_k^{(1)})$

↳ etiket değişmeye kadar devam ederiz.

$$\hat{\mu}^{(N)} = \frac{1}{N} \sum_{i=1}^{N} x_i$$



buna kestericinin varyansı

$$\mathcal{E}(d,\theta) = E[(d(x)-\theta)^2]$$ → varyans

$$b_\theta(d) = E\{d(x)\} - \theta$$

$$E\{m\} = E\left\{\frac{\sum_{t=1}^{N} x_t}{N}\right\} = \frac{1}{N}\sum_{t=1}^{N} E\{x^t\} = \frac{1}{N}\sum_{t=1}^{N} \mu = \frac{N\cdot\mu}{N} = \mu$$

$$Var\{m\} = Var\left\{\frac{\sum_{t=1}^{N} x_t}{N}\right\} = \frac{1}{N^2}\sum_{t=1}^{N} Var\{x_t\} = \frac{N\cdot Var\{x_t\}}{N^2}$$

$$= \frac{var[x_t]}{N} = \frac{\sigma^2}{N}$$

kestiricinin varyansı

$$s^2 = \frac{\sum_{t=1}^{N}(x_t - m)^2}{N} = \frac{\sum_{t=1}^{N}(x_t)^2 - Nm^2}{N}$$

$$E\{s^2\} = \frac{\left(\sum_{t=1}^{N} E[(x_1)^2]\right) - N\,E\{m^2\}}{N}$$

$$Var\{x\} = E\{x^2\} - E[x]^2$$

$$E[x^2] = Var\{x\} + E\{x\}^2 = \sigma^2 + \mu^2$$

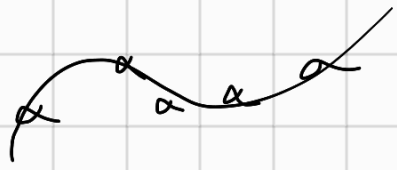
$$E[m^2] = \frac{\sigma^2}{N} + \mu^2$$

$$E[s^2] = \left(\sum_{i=1}^{N} \sigma^2 + \mu^2\right) - N\cdot\frac{\sigma^2}{N} - N\mu^2$$

↗ yanlı bi sonuç

$$= \frac{N\sigma^2 + N\mu^2 - \sigma^2 - N\mu^2}{N} = \frac{(N-1)\sigma^2}{N} \overset{?}{=} \sigma^2$$

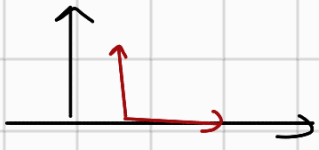
N ile çarpıp N-1 bölüp kestiriciyi değiştirsek çıkar



$$\sum_{i=1}^{N} C_i \left(e^{+jw_i k}\right) = f(k)$$

→ baz fonksiyonları

$$f(x) = \sum_{k=1}^{K} w_k \phi_k(x - x_k)$$



$$\langle f(x), g(x)\rangle \int_{-\infty}^{\infty} f(x)\, g(x)\, dx = 0$$

dik lik

Makine öğrenmesinde şart olmasa da olur base functions

haftaya Naive bayes